

Performance Testing

Date	01 October 2026
Team ID	SWTID-2026-8779
Project Name	PocketSmart AI – Your Smart Budget & Recommendation System
Maximum Marks	5 Marks

Performance Testing

- Tested the application for **response time and stability**.
- Verified planner pages and recommendation generation under normal usage.
- Checked API response and page-loading performance.
- Ensured smooth navigation between major modules.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Playwright
Type of Testing	Load Testing, Response Time Testing
Target Module / API	AI Recommendation API
Test Environment	Localhost
Test Date	01 october 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login	10	60 sec	Successful response
2	AI Recommendation Request	10	60 sec	Recommendation generated
3	History Retrieval	10	60 sec	History loaded successfully
4	Multiple API Requests	20	60 sec	System remains responsive

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.5 sec	Pass	Within target limit
2	Response Time (Max)	< 5 seconds	3.2 sec	Pass	Acceptable under load
3	Throughput (Req/sec)	< 20 seconds	15 req/sec	Pass	Good request handling
4	Error Rate	< 1%	0.5%	Pass	Very low errors
5	CPU Utilization	< 80%	65%	Pass	Within acceptable limit
6	Memory Utilization	< 80%	70%	Pass	Within acceptable limit

Step 4: Observations & Analysis

And screenshot

Key Findings:

System responded within the expected time under normal load.

Bottlenecks Identified:

AI API response time may increase during higher loads.

Optimization Steps Taken:

Reduced unnecessary API calls and optimized request handling.

Step 5: Screenshots / Evidence

